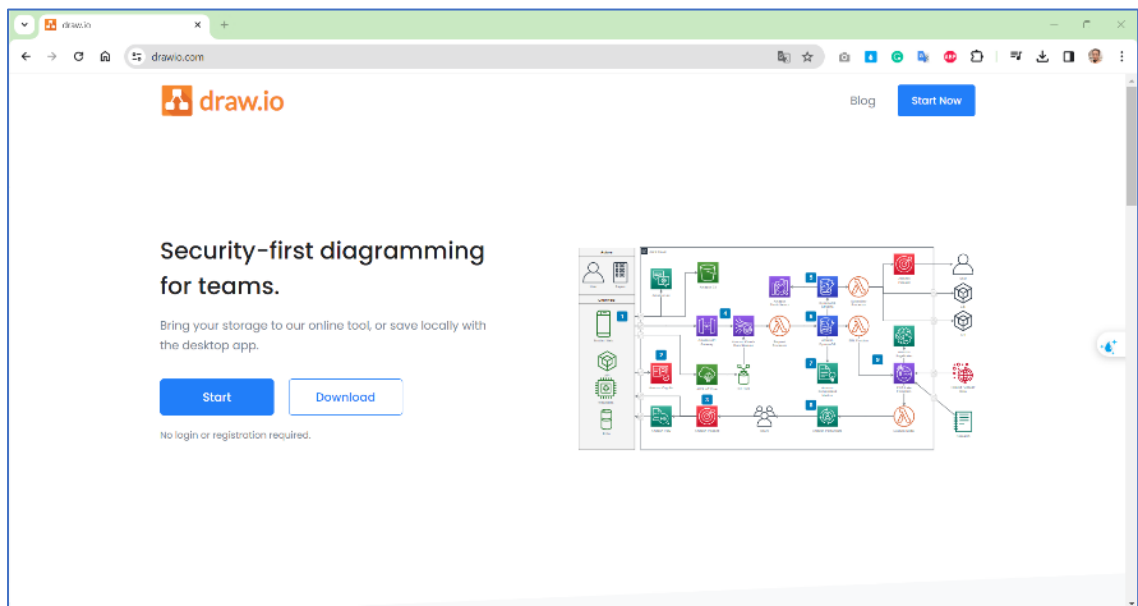


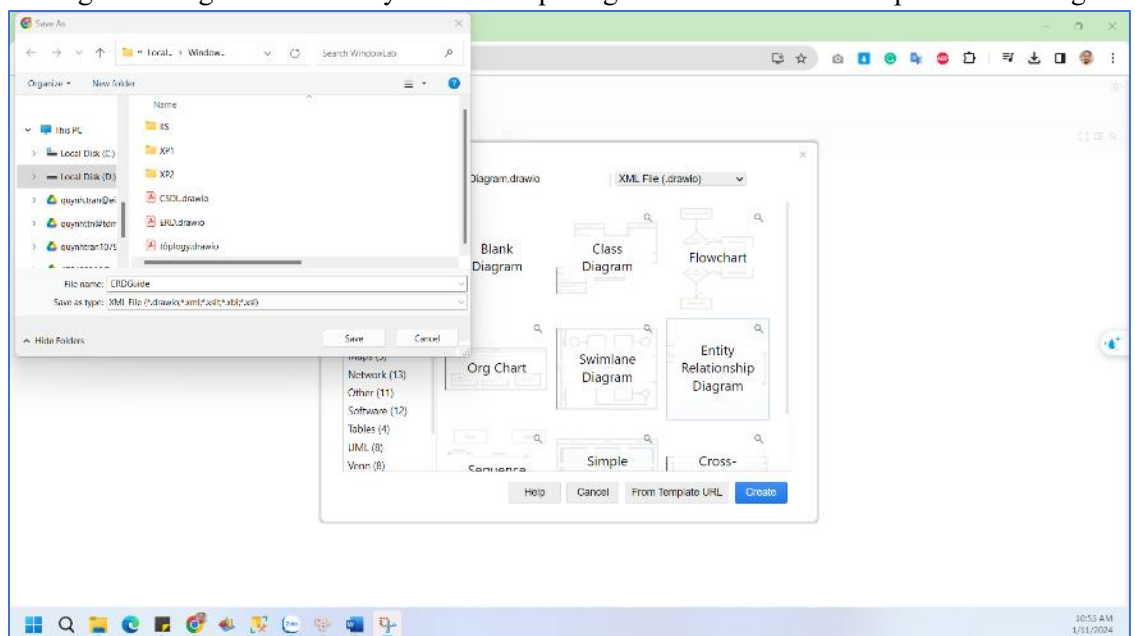
LAB 01: ENTITY RELATIONSHIP

I. Using Draw.io

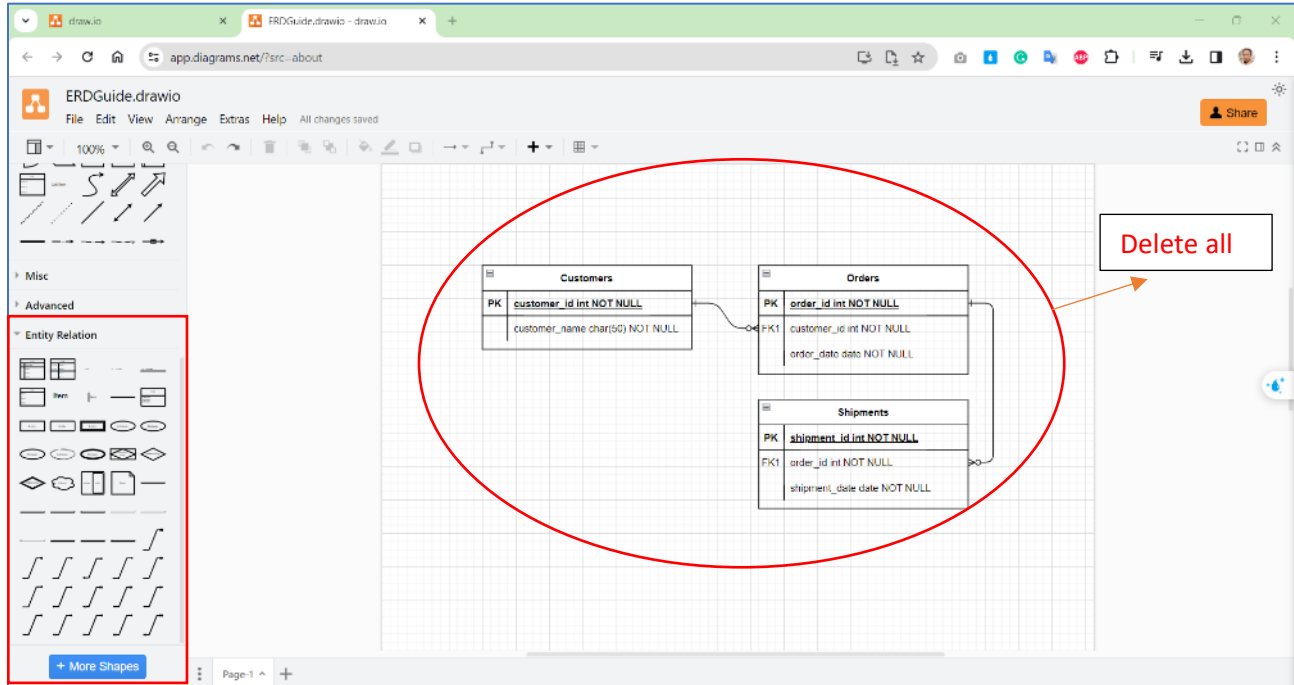
- You can access to <https://www.drawio.com/> then you can chose “Start Now” to use the online version or “Download” to use the offline version.



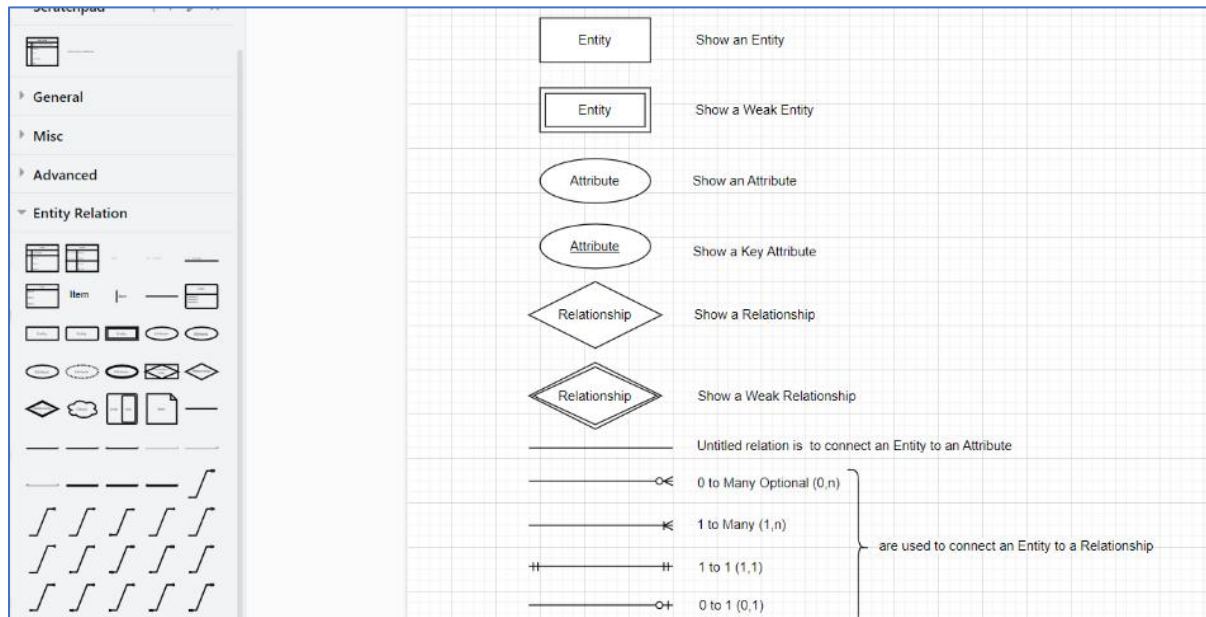
- Guiding to use online version for drawing ER diagram the following:
Chose “Creating new Diagram” -> “Entity Relationship Diagram” -> Create -> Set up name for Diagram -> Save



- Ctrl + A -> Deleting all of the things you don't use -> Finding in Left tab and Choosing "Entity Relation" – in this, including all of icon to draw ER diagram.



- Guiding to use icon for drawing ER Diagram:



II. Using draw.io to solve the problems below:

Guide:

1. Finding Entities, key Attributes and related Attributes
 2. Finding Relationships
 3. Finding weak Entities and weak Relationships (if any)
 4. Draw an ER model
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Problem 1. Building an ER model for a hospital

My hospital has many doctors working in different specialties. For each doctor, we need to keep basic information such as doctor code, full name, date of employment, and specialty. When a patient is admitted to the hospital, that patient is first received by one doctor, and each patient is associated with only one admitting doctor, while one doctor may admit many patients. After admission, a patient may be treated by one or more doctors depending on the medical condition. Some doctors may not treat any patients, while others may treat many different patients. In addition, the hospital needs to keep a full record of every time a doctor treats a patient, including the treatment date, treatment duration, and treatment result, so that we can conveniently monitor, manage, and follow up on the patient's care later.

Problem 2. Building an ER model for a University

The university needs to build an integrated system to manage all academic and student information in a consistent, clear, and efficient manner. In this system, the university has many faculties, each identified by a faculty code and faculty name; within each faculty, there are many departments, and each department manages many classes. Each class should store basic information such as class code, class name, and number of students. When a student is admitted to the university, the student will be assigned to a specific class. For each student, the system must keep complete information including student ID, full name, date of birth, gender, and address. Throughout the study period, students will take many different subjects, and after each subject, the university needs to record the student's exam score for that subject. In addition, at the end of the study program, each student will have a graduation score for academic management and evaluation purposes. For each subject, the system should store the subject code, subject name, and number of credits. Furthermore, each student is issued one library card, and each library card belongs to only one student; the library card information should include card number, issue date, and expiration date. The system must be able to manage all of these relationships completely in order to support the university's teaching, monitoring, and student administration activities effectively.

Problem 3. Building an ER model and relational data model for 1 hotel needs to store information as follows:

Our hotel needs a system to manage rooms, guests, services, and payment invoices in a clear and organized way. The hotel has many rooms, and for each room we need to store the room code, room name, the floor where it is located, and the room rates for both daily and weekly stays. Each floor also has its own code and name. When a guest comes to stay, we need to record the guest's identity card number, full name, and phone number. During the stay, the guest will stay in a specific room, and the hotel must issue an invoice that records the invoice code, guest information, room used, arrival date, departure date, and the room charge. If a guest stays for one or more full weeks and some extra days, the full weeks will be charged at the weekly rate and the extra days will be charged at the daily rate. In addition to room charges, the hotel also provides extra services such as laundry, and for each service we need to store the service code, service name, and unit price for one use. The invoice must also record the services used by the guest, the total service charge, and the final total amount to be paid, which includes both the room charge and the service charge.